

AI MOBILE APP FOR EARLY IDENTIFICATION OF STUDENT MENTAL HEALTH VIA BEHAVIOR & SPEECH

Project ID: 25-26J-278

Project Proposal Report

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August 2025

Student Burnout Detection and Intervention System using Machine Learning

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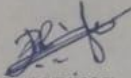
Declaration

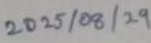
Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.


Signature of the supervisor:


Date

Abstract

Depression, anxiety, and stress are mental health issues and they are becoming highly widespread and prevalent among us college students, and to be quite honest, we tend to ignore them to such an extent that they are soon subject to a breakdown. Its common methods of checking in, which are: self-report or waiting until one visits the clinic, are patchy and occur only when the condition deteriorates. There is a major loophole on early and continuing monitoring. To address that, we are developing an AI-based mobile application to detect a sign of a descent before it can occur because it can analyze behavioral and emotional patterns of user activity, pick out Speech tone, phone screen usage, typing habits, sleep logs, and movements indicators.

Burnout is becoming a massive challenge to the students as well as the workforce who work part-time, primarily due to the academic strain, excessive hours as well as a hectic life. It spins out emotional burnout, experiences disidentification and devalues what you thought you made, which can smash against your mind, academic performance, and performance. Before we notice its insidious beginnings in moderate burnout, most of us start to think they are feeling tired. Losing such red flags may be followed by health issues, drop-in productivity, and increased likelihoods of abandoning a course or losing a job.

In order to assist with this, the Burnout Detection and Intervention System extracts behavioral and text data and applies machine learning to identify early burnout warnings. It analyses your behaviours based on NLP that browses using what you input and monitors default indicators such as screen time, sleeping habits and iPhone activity. The system offers you personalized help once burnout has been identified: relaxation exercises, lifestyle tips, or connections to a specialist.

The entire strategy is supposed to provide students and pros an instrument that is easy to use, trustworthy, and helpful to track their mental state, reduce burnout, and increase their productivity. The system remains close, suggesting the effectiveness of AI, and effective psychology frameworks in actual life.

Keywords: Burnout, Mental Health, Machine Learning, Intervention, NLP, Behavioral Data

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List of ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
LLM	Large Language Model
ASR	Automatic Speech Recognition
STT	Speech-to-Text
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
SVM	Support Vector Machine
MBI	Maslach Burnout Inventory
CBI	Copenhagen Burnout Inventory
PSS	Perceived Stress Scale
STAI	State–Trait Anxiety Inventory
HR	Heart Rate
HRV	Heart Rate Variability
EDA	Electrodermal Activity
EEG	Electroencephalography
ST	Skin Temperature
ROC	Receiver Operating Characteristic
PREs	Prediction Rule Ensembles

1. INTRODUCTION

The psychoanalysis stuff fellow, anxiety, depression, and stresses have become so much a part and parcel on campus. The entire grading, the new people, the social life and all are playing an insertive key to our psychological state. Although people are already concerned, it is still difficult to notice who is struggling quickly and provide assistance. The stale maneuvers, fill out the questionnaires or go through a psychologist interview, are subjective in nature, usually occurring on a few occasions and are too reactive. We contentedly require a kind of a thing that puts a check on us at all times, an impending check-in.

Even listening to how we speak, one may tell a lot about our mood. When you feel down or wired, then you may find your voice being flat and slowed down with a strange pause and little variation of pitch. All that could be picked up on by a phone-app in real time as you chat on a call or simply jotting your thoughts, and provided a snapshot of how you are progressing. In addition, other behaviors, such as the amount of screen time you receive or based on how you type, can indicate trouble in its infancy. An excessive or moderate use of the phone may also suggest that either we are closing down or we are nervous, and slowness with typing or a slew of corrections or derogatory words may be evidence of a change in our thought or mood.

Sleep is also a big deal. Monitoring your pre-sleep phone activity and the immediate post-sleep phone activity helps identify abnormal or disrupted sleep, which is, in turn, related to anxiety and depression. When we add all those data-points, including Speech, screen time, typing, sleep, we get a complete mental-health picture. The app does not diagnose but can raise red flags, provide you with self-care advice, and connect you with professional help, to help de-stigmatize or combat difficulty in getting care.

We are creating a work that utilizes AI and phones to provide special attention to detecting the signs of mental health before you even identify it. It is attempting to absorb small changes in your behavior or mood using time-series technology and deep-learning to

identify the large stuff before it appears. And it is based on privacy-first that keeps it accountable enough to early intervene so we and the campus health team can step in.

When we stop merely responding when someone throws the door open and instead start handling the problem before it grows to be an issue will allow us to have early solutions to problems and you will have more interventions that will allow us to vibrate better as a student body. We are just talking all about keeping us psychologically well in college.

1.1 Background and Literature survey

Mental health challenges are increasingly recognized as critical concerns in both academic and workplace environments. Students, in particular, face high levels of stress, academic workload, emotional exhaustion, and social isolation, which can progress into severe conditions such as burnout, depression, or anxiety if not identified and managed at early stages [1], [2]. Traditional approaches to mental health monitoring rely on self-report questionnaires, clinical assessments, or counselor evaluations, which are effective but often time-consuming, costly, and inaccessible for many individuals, leading to under-diagnosis of mental health risks [3].

Recent advancements in artificial intelligence (AI), natural language processing (NLP), and behavioral analytics have enabled automated systems that can detect subtle mental health indicators by analyzing language use, speech patterns, and behavioral data [4]. Such technology-driven tools act as preventive systems, offering early detection, personalized feedback, and interventions before conditions worsen [5].

Among mental health conditions, burnout has gained significant attention. The World Health Organization (WHO) defines burnout as an occupational phenomenon caused by chronic workplace or academic stress that has not been successfully managed [6]. Burnout manifests through emotional exhaustion, depersonalization, and reduced productivity, and has been directly linked to increased risk of dropouts, absenteeism, decreased performance, and long-term psychological illness [7], [8]. Despite its significance, detecting burnout early is challenging, as it develops gradually and is typically identified only after symptoms have become more pronounced [9].

To address these limitations, our proposed AI-based Burnout Detection and Intervention System is integrated as a function within the larger project, the AI Mobile App for Early Identification of Student Mental Health via Behavior & Speech. This system combines multiple components:

Textual Analysis – Leveraging transformer-based NLP models (e.g., BERT, GermanBERT) to identify linguistic markers of burnout in journals, logs, or chat inputs [10].

Behavioral Analysis – Monitoring digital engagement patterns such as screen time, task completion rates, and activity levels to identify signs of emotional exhaustion [11].

Standardized Questionnaires – Validating AI predictions against clinically recognized instruments such as the Maslach Burnout Inventory (MBI) and the Copenhagen Burnout Inventory (CBI) [12], [13].

Personalized Interventions – Providing evidence-based recommendations such as micro-breaks, mindfulness exercises, task management strategies, and counselor referrals [14].

By delivering these capabilities via a cross-platform mobile application, the proposed system offers a real-time, accessible, and cost-effective tool that goes beyond detection to support recovery actively. Unlike conventional methods, this solution functions as a proactive mental health companion, empowering students to recognize early warning signs of burnout and adopt healthier coping mechanisms.

There are lots of studies that have been conducted about human depression, anxiety, and stress. The following studies specifically address the Burnout Detection and Intervention System.

One such study, titled "Using Natural Language Processing to find Indication for Burnout with Text Classification: From Online Data to Real-World Data" was conducted in 2024 by Mascha Kurpicz-Briki, Ghofrane Merhbene, Alexandre Puttick, Souhir Ben Souissi, Jannic Bieri, Thomas Jörg Müller, Christoph Golz. The primary objective of this study was to investigate the capability of transformer-based models, specifically GermanBERT, to detect textual indicators of burnout by leveraging multiple data sources. The models employed comprised four fine-tuned GermanBERT classifiers trained individually on (1)

online burnout-related data, (2) a manually curated dataset (BurnoutExpressions v1), (3) a synthetic dataset extended with GPT-generated sentences (BurnoutExpressions v2), and (4) a combined dataset integrating online and synthetic corpora. The training parameters utilized included an 80/20 train-validation split, 2–3 epochs of fine-tuning, batch sizes of 16 (training) and 64 (evaluation), and a weight decay of 0.01 to optimize generalization. The experimental results demonstrated accuracies stabilizing around 0.83–0.85 for the BurnoutExpressions v2 and combined datasets during evaluation. At the same time, real-world testing yielded F1-scores up to 0.61, underscoring both the promise and limitations of applying NLP models trained on non-clinical data to authentic clinical contexts.[15]

One such study, titled "Detecting Burnout: Identifying Key Symptoms Using Standard and Machine Learning Methods," was conducted in 2023 by Tavella G., Spoelma M., Parker G. The primary objective of the study was to evaluate whether a newly developed 34-item burnout measure could effectively distinguish between individuals who self-identified as experiencing burnout and those who did not. To achieve this, the researchers employed both statistical models, such as receiver operating characteristic (ROC) analyses, and machine learning models, specifically prediction rule ensembles (PREs), to test classification performance. The main parameters used included participant responses to the 34 burnout-related items, which were coded on Likert scales or dichotomized into the presence/absence of symptoms. The analyses demonstrated good discriminatory capacity, with ROC-derived cutoff scores yielding overall classification accuracies of 73.1% (nondichotomized data) and 72.5% (dichotomized data). In contrast, the machine learning models slightly outperformed these results, achieving accuracies of 73.76% and 75.20%, respectively.[16]

Another study, titled "Using Machine Learning in Burnout Prediction: A Survey" was conducted in 2021 by Grządzielewska M. The primary objective of the reviewed studies is to enhance the prediction of occupational burnout by utilizing machine learning techniques, which can handle complex, non-linear relationships and large, multi-source datasets, such as questionnaires, digital sensors, and health records. Various models have been applied, including supervised learning methods like decision trees, support vector machines (SVM), convolutional neural networks (CNN), and multitask learning, as well as unsupervised

approaches like k-means clustering. Commonly used parameters include sociodemographic characteristics, work-related factors (such as workload, autonomy, and role stress), psychological traits (such as neuroticism and self-efficacy), and physiological stress indicators. Reported accuracies vary depending on the dataset and algorithm; for example, CNN models achieved high accuracy in predicting nurse burnout, while logistic regression performed competitively in some comparisons, and clustering-based approaches demonstrated around 70% accuracy in early burnout detection.[17]

Another study, titled "Students' Burnout Symptoms Detection Using Smartwatch Wearable Devices: A Systematic Literature Review" was conducted in 2025 by Paschalina Lialiou and Ilias Maglogiannis. The primary objective of the reviewed studies is to utilize smartwatch wearable devices integrated with AI-driven predictive models for the early detection of burnout symptoms, particularly stress, among students. These studies employ various models, including random forest, k-nearest neighbors, XGBoost, deep learning techniques, and temporally hierarchical attention mechanisms, to analyze physiological and behavioral data. The most frequently used parameters are heart rate (HR), heart rate variability (HRV), electrodermal activity (EDA), skin temperature (ST), and electroencephalography (EEG) signals, often complemented by self-reported stress and anxiety scales such as the Perceived Stress Scale (PSS) and State–Trait Anxiety Inventory (STAI). Reported accuracies vary depending on the models and input modalities, ranging from a mean absolute error (MAE) of 5.87 bpm for HR monitoring with the Fitbit Versa 2 to classification accuracies of 99.98% using EEG-based features and 99.51% using multimodal physiological signals, such as EDA, HR, and ST.[18]

1.2 Research Gap

There is a wide range of systems designed to identify stress, anxiety and depression using questionnaires, wearable gadgets or speech recognition. A few scientists have used machine-learning and natural language processing technology to detect burnout symptoms

using textual and survey data. However, the vast majority of the current systems do not feature any form of personalised and real time interventions; they only detect and do not exist in any form of interventions.[19]

As an example, transformer-based models, including GermanBERT, were used in a study to detect burnout language patterns on synthetic data in approximately identical accuracy; nevertheless, when exposed to real data, somewhat widely, performance declined markedly. Although this is a sign of potential, absence of generalizability restricts its clinical application. Similarly, a different study provided a new 34-item questionnaire of burnout which effectively differentiated individuals with burnout and burnout-free individuals. However, it involved Survey Responses that needed to be completed manually and did not include any behavioral data and automatic feedback.[20]

A number of research works used supervised and unsupervised machine-learning frameworks, such as convolutional neural network, support vector machine, and clustering type, to predict occupational burnout through the analysis of health records and questionnaires. Although these methods have become satisfactorily accurate, they were heavily reliant on structured clinical data, which makes them a bit less relevant in an academic environment due to the more relevant nature of real-time behavioral data like screen time or study activity. Wearable-based detectors with physiological measurement (HR, HRV, EEG) were also suggested in other studies and achieved extremely high rates of accuracy; the fact that wearables are a requirement makes these systems expensive, not scalable, and unfitting large groups of students.[21]

Conversely, the presented system correlates textual evidence, e.g. mood notes and journals, with behavioral outcome, e.g. screen time, number of tasks performed and number of studies approached and engaged in, to determine burnout levels. Further, it, unlike other systems, features a system that includes a reinforcement-learning agent to provide personalized interventions (e.g. study-breaks, meditation, reorganization of work), consequently eradicating the disparity between detection and action.

Application Reference	Burnout Detection (Text-based/NLP)	Burnout Detection (Survey/Questionnaire)	Burnout Detection (Wearables/Physiological Data)	Personalized Intervention
Research [15]	✓	X	X	X
Research [16]	X	✓	X	X
Research [17]	✓	✓	X	X
Research [18]	X	X	✓	X
Proposed System	✓	✓ (optional validation)	X	✓

1.3 Research Problem

Burnout has become one of the most critical mental health issues in both academic and work settings. Students and employees are increasingly experiencing symptoms like emotional exhaustion, detachment, and decreased productivity caused by heavy workloads, academic stress, and neglecting self-care. Often, burnout isn't addressed until it becomes severe, resulting in long-term health problems and lower performance. Unlike stress or anxiety, which can be temporary, burnout is a persistent condition that needs early detection and treatment to avoid getting worse.

Currently, burnout is primarily identified through manual questionnaires and self-reports, such as the Maslach Burnout Inventory (MBI) and the Copenhagen Burnout Inventory (CBI), or through ad hoc surveys conducted by institutions. While these tools work well in structured research settings, they are time-consuming, subjective, and not scalable for real-time monitoring. As a result, many cases of burnout go unnoticed until they become severe, impacting academic performance, physical health, and overall well-being. Students often

underreport their struggles because of stigma or lack of awareness, which makes the problem worse.

In Sri Lanka and many developing regions, burnout detection and intervention systems are almost non-existent, especially for students. Most universities and organizations rely solely on periodic surveys or counseling sessions, which do not capture real-time behavioral changes. Students, in particular, face unique challenges, including intense exam pressure, prolonged study hours, irregular sleep patterns, excessive screen time, and a lack of physical activity—all of which contribute to burnout. A preliminary survey conducted among undergraduates in Sri Lanka revealed that more than 70% of students experience symptoms of fatigue, lack of motivation, and reduced concentration. Still, only a small fraction seek help from mental health professionals due to stigma, lack of accessibility, or unawareness.

The existing systems reviewed in the literature demonstrate that while AI models (such as BERT-based text analysis, CNN models on occupational burnout data, and wearable-driven detection) have made progress, they still suffer from critical limitations. Most systems either (1) focus on text data alone, (2) rely heavily on expensive wearable devices, or (3) stop at detection without providing personalized interventions. These constraints make them challenging to deploy widely, especially in low-resource academic environments.

Therefore, the need for an innovative, scalable, and cost-effective system to detect burnout in real time is evident. Unlike existing models, the proposed system will combine textual indicators (such as vague, negative, or emotionally drained language in journal entries or mood logs) with behavioral data (including screen time, irregular sleep patterns, and reduced physical activity) to provide a comprehensive view of student well-being. Furthermore, the system will go beyond detection by integrating a machine learning (ML)-based intervention mechanism that suggests breaks, mindfulness activities, and time management tips tailored to individual needs.

In short, burnout not only affects the mental health of students but also has a direct impact on their academic performance, employability, and long-term productivity. If left unaddressed, it results in economic losses for institutions and compromises the overall quality of education. Since there are no comprehensive, AI-driven mobile applications

currently available in Sri Lanka to address this issue, the proposed Burnout Detection and Intervention System will fill a critical research and practical gap by offering an innovative solution that is real-time, personalized, and accessible.

2. OBJECTIVES

2.1 Main Objectives

The main purpose of the research is to create an AI-powered module, which can anticipate the hypothesised risks of a mental decline in the future by processing multimodal behavioural and emotional data interactions such as the tones of the voice, sleep habits, the way screens and keyboards are being used, etc. Using temporal deep-learning models based on such continuously received signals, the system is expected to determine specific personalised windows of vulnerability days or weeks before clinical symptoms appear. By providing timely and interpretable alerts and risk prediction, the solution will empower students to take positive self-care steps before crisis development, and it will also allow healthcare providers to act before escalation of lung problems, thus transforming mental-health care delivery into proactive assistance, rather than, reactive diagnosis.

2.2 Specific Objectives

1. To analyze student-generated text (mood logs, journal entries, academic reflections) using NLP for burnout-related language patterns.
2. To monitor behavioral data (screen time, task completion rates, engagement patterns, study duration) for early burnout indicators.
3. To develop a classification model capable of categorizing students as “at risk of burnout,” “neutral,” or “healthy.”
4. To design a machine learning (ML)-based intervention model that delivers personalized support (e.g., study break reminders, motivational prompts, mindfulness activities).
5. To validate the system against standardized questionnaires.

3. Methodology

The advanced system suggested can identify early symptoms of burnout in the students and professionals. Also, the system is used to categorize the burnout intensity, thus creating awareness and support that offers users in time, thus helping to keep their mental health in a healthy state in terms of timely prevention and protection measures. First, the system receives input data on the part of its users which involves the text data (including journal, reflections and chatlogs) and behavioral data (including screen time, sleep patterns, and activity levels). This information is then preprocessed and processed using techniques of natural language processing (NLP) and machine-learning models to obtain burnout-related indicators. Lastly, the severity of burnout is unified into levels of low, moderate or high risk. According to this classification, the system will suggest the interventions which are dependent on the severity identified with i.e. mindfulness practice, lifestyle change, or contact with the professional counseling service.

3.1 System Architecture

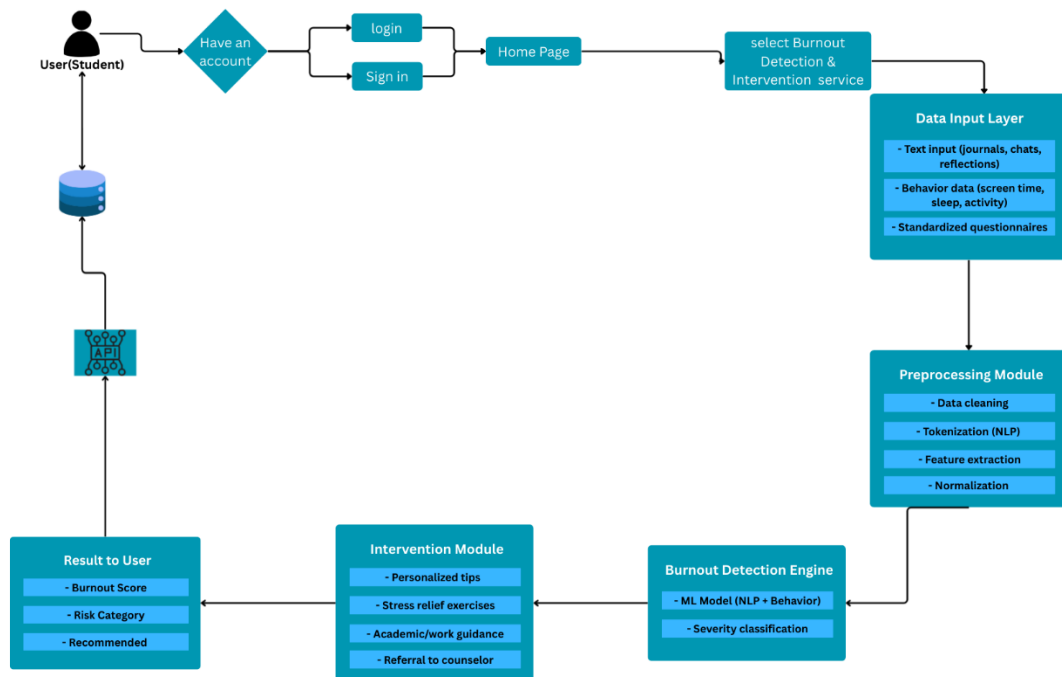


Figure 1.1

Figure 1.1 presents a high-level schematic of the suggested burnout detection and intervention component. The system first engages the student through a mobile program, whereby the user can input text logs or mood records. At the same time, the app is constantly gathering behavioral metrics, including the activity on screen, a task-completion history, and the duration of usage of the app.

The acquired textual and behavioral information are preprocessed then sent to a backend server. Afterward, machine-learning and natural-language-processing models are used to evaluate the processed data and determine the degree of burnout in the student.

The intended component will contain three sub modules, as shown in Figure 3.2:

1. Burnout Detection (Text) Use Because this model is used to identify instances of burnout indicative language in student generated text (i.e. presence of expressions such as exhaustion, disengagement etc.).
2. Burnout Detection (Behavior) – Machine-learning models analyze the behavioral evidence (slowing the speed of task completion, increasing screen-time).
3. Severity Assessment / Intervention- A fusion model integrates the textual and behavioral results, rate burnouts severity (low, medium, high) and a reinforcement-learning agent suggests an individual approach to interventions.

Table 1: Technologies, Techniques, Architectures, and Algorithms.

Category	Choices Used
Technologies	Python, TensorFlow Lite, PyTorch Mobile, React Native, Flask/FastAPI
Techniques	Natural Language Processing (NLP), Machine Learning (ML)

Algorithms	BERT/DistilBERT for NLP, Random Forest & SVM for behavior analysis
Architectures	Transformer-based models (BERT), Fusion Ensemble Model

3.2 Software Solution

The proposed solution will follow the **Agile methodology** with **Scrum** to ensure iterative development and continuous feedback.

Development Phases:

- **Requirement Gathering:** Collection of behavioral and emotional datasets, including speech tone, screen usage, and sleep patterns; surveys with students for baseline data.

- **Feasibility Study (Planning):**

- Economic Feasibility

The system under consideration can be imagined as a mobile application that will be able to work effectively without the need to create expensive infrastructure. Together with the open-source Launchpads, it will be built on lightweight machine-learning models, thus serving to make it cost-effective all the way to development and deployment.

- Scheduled Feasibility

Sub -component in focus bears direct impact on the success of the project and should be done within the specified time limit. The various developmental milestones such as the data collection, training models, application development, testing will be aligned with the academic calendar to ensure a timely delivery of the completed artefact.

- Technical Feasibility

The development of the solution front-end involves knowledge in the field of mobile application development, the field of natural language processing, machine learning, and cloud-based API proficiency to successfully implement the solution. The technical feasibility is supported by the open-source tools available such as TensorFlow (PyTorch, React Native, Firebase, and others), all of which, together, provide a facsimile base according to which the evolution is possible.

- **Implementation (Development)**

- The development stage will constitute the following sub-functionalities:
- Design of the mobile application based on React Native to allow cross-platform deployment.
- The scheme where burnout is identified based on textual information, achieved using an NLP-wise classification model.
- Behavioural pattern analysis using machine-learning and statistical methods.
- Severity classification into low, moderate, and high burnout risk categories.
- The supply of a recommendation system to include customised interventions, e.g. mindfulness meditation exercises, digital-detoxing warnings, or links to professional counselling.

- **Testing:**

There will be testing activities which will ensure that the system meets all stated functional and non functional requirements, which errors, bugs, and performance are detected. Testing will involve a series of staged testing, unit testing, integration testing, system testing and acceptance testing whereby the end acceptance testing will be done by the actual users, who in this case, consist of students and professionals.

3.3 Commercialization

This study aims to create an early diagnosis and intervention mobile application on student burnout. The purpose audience of the proposed system is for students who are interested in higher education and want to know what subjects to pursue at university, guidance students and institutions that take special care of student welfare. The system will come in two versions: standard version and premium version. The basic version will include tools like: self-assessment questionnaires, risk level for burnout detection, tracking progress and overall self-care recommendations. The premium product offering will include higher level features such as focused intervention plans for students, AI-based behavioral analysis based recommendations for Wendt and more customised advice, optional access to a counselor (with student consent), and a feature for advanced analytics on progress.

In terms of accessibility, the app is designed ensuring that it is easy to use and no advanced technical knowledge is required. Students of all age groups and fields can use the system very easily in a short training. Our team is made up of technically strong individuals majoring in bachelor's in IT and software engineering skills who have had experience along with mobile application development and AI. The estimated annual spend with respect to development and maintenance is around LKR 50,000 (US \$150) per month.

On the marketing side, the product can be marketed via university-level awareness programs, student societies, wellness sections, live-free forums etc. and through various digital platforms like social media. From a business perspective, the solution can be financially self-sustaining and scalable by offering the solution on a freemium model, institutional subscription, and in-app premium add-ons to universities domestically and internationally.

4. Project requirements

4.1 Functional Requirements

1. Burnout Detection:
 - NLP-based text analysis for burnout-related language.
 - Behavioral monitoring (task completion, screen usage, procrastination index).
2. Real-Time Feedback:
 - Generate alerts when burnout indicators cross thresholds.
3. Interventions:
 - Personalized interventions (study break prompts, motivational nudges, mindfulness exercises).
4. User Interface:
 - Mobile app with a dashboard for tracking engagement and well-being trends.

4.2 Non-functional Requirements

1. Availability

The system should always be accessible to both the students and the staff since the systems of burnout can manifest themselves at any point. Cross-linked to a cloud deployment, along with an automated backup process, ensures that the mobile application as well as back-end services are always available, therefore reducing the amount of operational downtime.

2. Performance

Application: This is expected to demonstrate quick response times and the average latency of less than two seconds on submitting textual and behavioral data and less than five seconds to generate burnout risk assessments. Inference machine-learning, such as

classification and recommendations, have to be scaled down to be mobile with no or limited battery power and processing consumption.

3. Reliability

The system must ensure there are uniform results during normal and peak finding use cases. To take one example, the burnout score and recommendations should be consistent relative to the same input of the user. The validation framework that will support the reliability of the results will be the use of standardized burnout questionnaires (MBI/CBI).

4. Scalability

The architecture has to support thousands of simultaneous users across different universities without being obviously affected by the usage. These require a hyper-scalable server architecture—with load balancers and microservice elements—as well as decoupled machine-learned models that can be re-trained on larger datasets.

5. Usability

Interfaces should be easy to use and simple to understand as students should form moods, behaviours, and journal entries with a minimum amount of effort. The system should be expressed in recommendations that are simple, understandable language on how to act based on the results.

6. Accessibility

The system has to be available to students with disabilities. Additional features consist of text-to-speech capabilities, multilingual support (Sinhalese, Tamil, English), and a high-contrast graphics to enable use by visually discriminated students, hence inclusivity among students of varied profiles.

4.3 System Requirements

1. Hardware Requirements

- a. Client side (student devices): Any Android smartphone with at least 3GB RAM, a dual-core processor, and internet connectivity.
- b. Server side: Cloud-hosted servers (AWS, Azure, or GCP) with GPU support for ML model inference.

2. Software Requirements

- a. Frontend: Flutter for cross-platform mobile app development.
- b. Backend: Node.js/Express.js for API services.
- c. ML services: Flask/FastAPI for hosting burnout detection models.
- d. Other tools include GitHub, Docker for containerization, and CI/CD pipelines for deployment.

3. Database Management

- a. Database system: SQLite for structured data for user logs and questionnaire results.
- b. NoSQL: MongoDB/Firestore for unstructured behavioral and text data.
- c. Data security: End-to-end encryption, role-based access

4. Machine Learning Model and Techniques

- a. Text Analysis: Transformer models fine-tuned for burnout-related language patterns.
- b. Behavioral Analysis: Random Forests, XGBoost, or LSTMs to model time-series behavioral patterns (screen time, activity levels).
- c. Fusion Layer: Late fusion to combine text + behavior features into a final burnout risk score.
- d. Intervention Module: Suitable machine learning model to recommend intervention.

- e. Validation: Standardized burnout questionnaires used for ground-truth validation.

4.4 User Requirements

This mobile application will be developed for Student user groups:

Students - students can see their burnout risk levels with easy proactive alerts and access to self-care suggestions for easier burnout prevention than if a more severe decline only occurs. They can also act as a reference to their progress over time via weekly reports, and show reminders to maintain a healthy balance between studying and life.

5. Grantt Chart

Month	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec-Jan	Feb	Mar	Apr	May
Brainstorming Session														
Topic Selection														
Supervisor and Co-supervisor selection														
Group Registration														
Create TAF Doc														
TAF Submission														
Create Log Books														
Create Proposal Presentation														
Proposal Presentation														
Create Proposal Report														
Proposal Report Submission														
UI Designing														
Implementation of work														
Training Data Sets														
Working on ML Algorithms														
PP1 Creation (Progress Presentation 1)														
PP1														
Writing Research Paper														
Research paper submission														
PP2 Creation (Progress Presentation 2)														
PP2														
Create Final Report														
Final Report Submission														
Testing the application														
Create Project Status Document														
Create Final Presentation														
Final Presentation and VIVA														

5.1 Work Breakdown Structure (WBS)

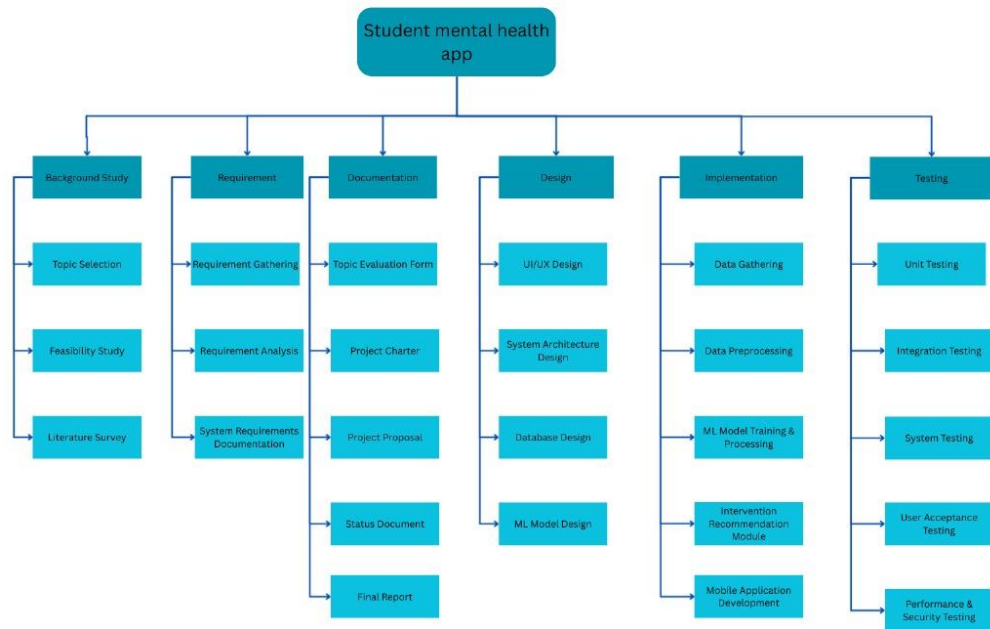


Figure 2

6. BUDGET AND BUGET JUSTIFICATION

Table 6.1 provides an overall budget for the overall proposed system.

Expenses	Requirement Cost (Rs.)	Justification
Travelling cost for surveys and awareness programs	15,000.00	To conduct initial data collection, distribute questionnaires, and raise awareness about burnout among students at universities.
Data Collection (standardized questionnaires, survey tools, digital forms)	10,000.00	Required for acquiring reliable datasets from students for training, testing, and validating the system.
Deployment cost (server hosting, backend services)	10,796.00 / month	To maintain system availability and reliability through cloud-based deployment.
Hosting cost in Play Store	8,076.00	One-time cost to publish the mobile application on the Google Play Store for student access.
Miscellaneous (printing, promotional activities, student awareness sessions)	5,000.00	For marketing, leaflets, and awareness campaigns to ensure successful adoption of the app.
Total Expenses	48,872.00 + 10,796.00/month	—

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